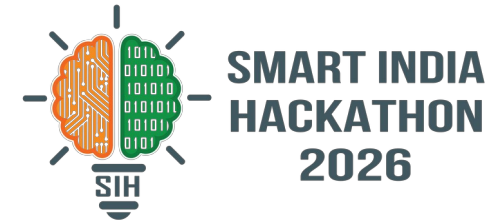


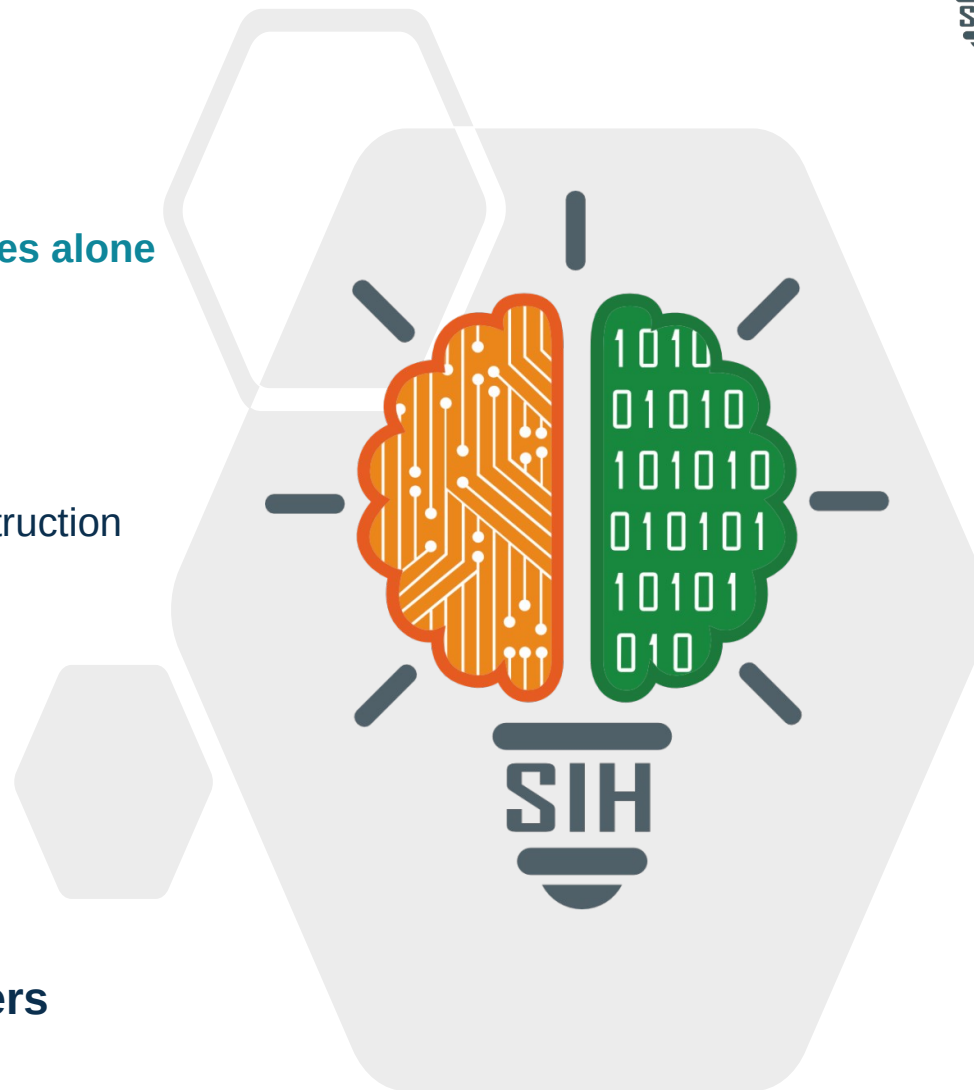
SMART INDIA HACKATHON 2026



OceanSight

Seeing 1000 m beneath the ocean surface — from satellites alone

- **Problem Statement ID – SIH26066**
- **Problem Statement Title – OceanEmbed – Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations**
- **Theme – Space Technology**
- **PS Category – Software**
- **Team ID – 135494**
- **Team Name (Registered on portal) – CodeCrafters**



OceanSight

Daily 0–1000 m ocean temperature for the North Indian Ocean — from satellites alone



Proposed Solution

Detailed explanation of the proposed solution

OceanSight reconstructs ocean temperature at **15 depths (0–1000 m)**, **every day**, on a **0.25° grid** over the Bay of Bengal & Arabian Sea — using only **5 satellite surface fields** — and turns it into cyclone-heat and mixed-layer products for ocean analysts and forecasters.



How it addresses the problem



THE GAP

Floats, buoys & ships sample the subsurface only at scattered points and times



OCEANSIGHT

Satellites see the whole basin daily; a deep network learns the physical surface → subsurface link (e.g. sea-level anomaly → thermocline depth)



Satellite inputs
SST · SSS · SLA ·
currents · winds



Harmonise
QC + regrid to
0.25°, daily



Satellite embedding
U-Net encoder
compresses the basin



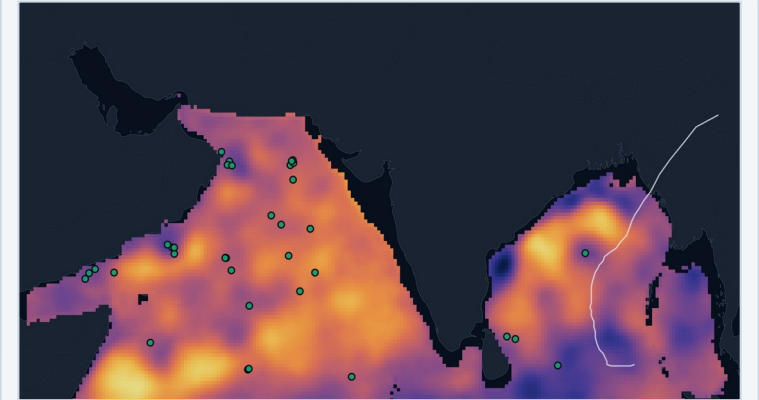
Reconstruct
15 depths, 0–1000 m
+ uncertainty



Derive
TCHP · MLD ·
D20 · D26



Deliver
GIS dashboard ·
API · reports



Working prototype: reconstructed temperature at 100 m on 11 May 2023 (held-out year) — dark purple ≈ 17 °C to yellow ≈ 28 °C; green dots = Argo floats; white line = Cyclone Mocha track (IBTrACS)



INNOVATION AND UNIQUENESS OF THE SOLUTION — why OceanSight is different



USP 01

Validated beyond training data

Scored against real Argo floats from a year it never saw — next to the training product's own error ceiling and an independent EN4 cross-check.

2,639 held-out Argo profiles (2023)



USP 02

Calibrated to the real ocean

Every profile carries an error bar calibrated on 2022 floats that holds on 2023 — analysts can see when not to trust a value.

70% of Argo values within $\pm 1\sigma$ (ideal 68%)



USP 03

Reconstruction → decisions

Daily cyclone heat potential, mixed-layer and isotherm depths at 0.25°, replayed on real IBTrACS cyclone tracks.

TCHP · MLD D20 · D26 — every day, 2019–2023

Satellite surface fields → validated, uncertainty-aware 3-D ocean temperature → INCOIS decision products

Technologies to be used

Satellite data NOAA OISST (SST) · SMAP + SMOS (SSS) · NOAA altimetry (SLA, currents) · NCEI Seawinds

Target & validation HYCOM 1/12° analysis (GLORYS adapter built) · Argo via argopy · Met Office EN4

ML / AI PyTorch U-Net (satellite embedding) · LightGBM baseline

Data engineering Python · xarray · Zarr (daily cubes) · NumPy regridding

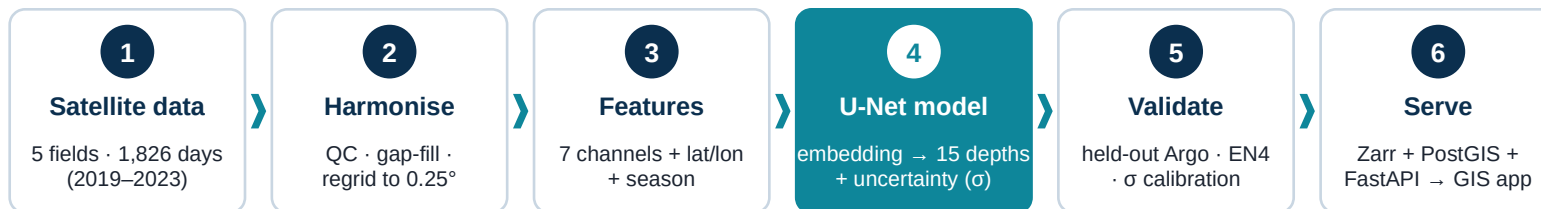
Geospatial / DB PostgreSQL + PostGIS (spatial indexes, nearest-float queries)

Backend FastAPI — typed REST API, PDF/CSV reports

Frontend / GIS Next.js · MapLibre GL · deck.gl · Recharts

Engineering Docker · GitHub Actions CI · 44 automated tests · CPU-only

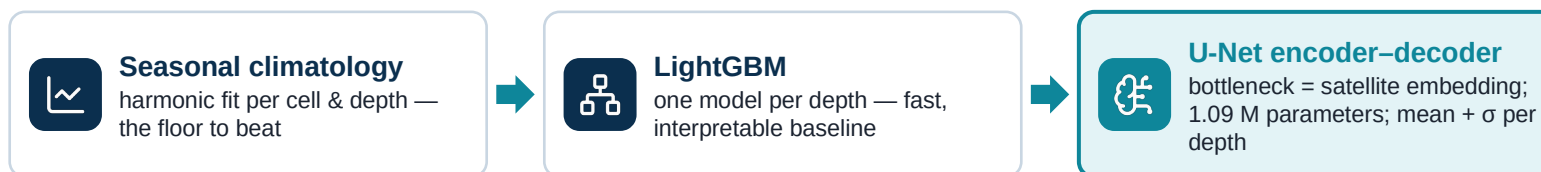
Methodology and process for implementation



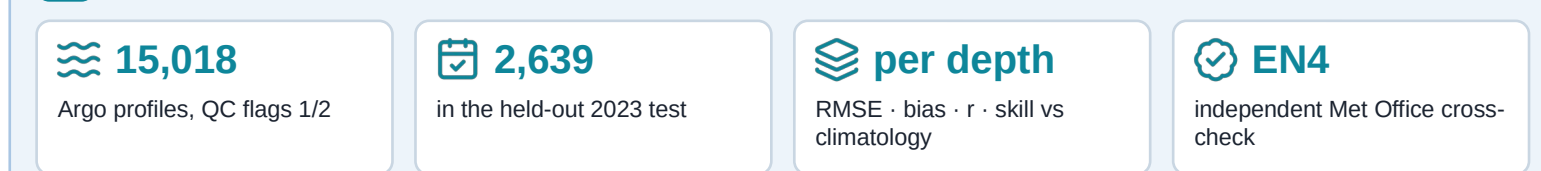
Honest evaluation — split by whole years, never random days



Models, compared side by side — the deep model has to earn its place



VALIDATION — against the real ocean, not just the training data



 Analysis of the feasibility of the idea working prototype, validated on the held-out year 2023 **2,639**

independent Argo profiles from 2023 — a year the model never trained on

 **0.89 °C**

mean temperature error (RMSE, 15 depths) vs 1.10 °C for climatology

 **1.31 °C**

error at 100 m (thermocline) vs 1.80 °C for climatology

 **70%**of Argo values within $\pm 1\sigma$ after calibration (ideal 68%; raw 41%) **FEASIBLE BECAUSE****Data****HIGH**

Every input, target and validation source used is open; Copernicus/ERA5 need only free sign-up

**Technical****HIGH**

Built and running end to end: ingestion → models → API → dashboard

**Compute****HIGH**

U-Net (1.09 M params) trained on a laptop CPU — no GPU

**Cost****₹0**

No paid data or services used

IMPLEMENTED

pipeline · models · validation · API · dashboard · Docker

PLANNED

hosted deployment · re-run on Copernicus/GLORYS with credentials

 Potential challenges and risks

Training target (a reanalysis) assimilates Argo → validation not fully independent

Raw model uncertainty was over-confident (41% of values within $\pm 1\sigma$)

Satellite salinity is gappy; removing it showed no measurable change in skill



GLORYS / Copernicus data need an account



Thermocline (20–150 m) is the hardest layer to reconstruct

 Strategies for overcoming these challenges

Hold out a whole year; show the target product's own error as the ceiling (0.79 °C); cross-check against EN4

Per-depth calibration fitted on 2022 floats, verified on 2023 → 70% within $\pm 1\sigma$, 94% within $\pm 2\sigma$ 

Merged SMAP + bias-corrected SMOS (absent days 209 → 1); re-test with Copernicus SSS



Credentialed adapters already built — switching source is a config change, no code change



Report error per depth, never one number; the uncertainty band widens where skill drops

Potential impact on the target audience



Ocean analysts & forecasters
INCOIS-style



Cyclone forecasters
heat available to storms



Fisheries-advisory analysts
mixed layer & thermocline



Ocean researchers
daily 3-D record

Decision support — not a replacement for operational ocean models

Benefits of the solution social · economic · environmental · operational

SOCIAL

Basin-wide cyclone heat potential (TCHP) and mixed-layer maps for every day and every 0.25° cell of the domain.

ECONOMIC

Built only on free satellite data; the model trains on a CPU; products are precomputed, so viewing needs no heavy compute.

ENVIRONMENTAL / CLIMATE

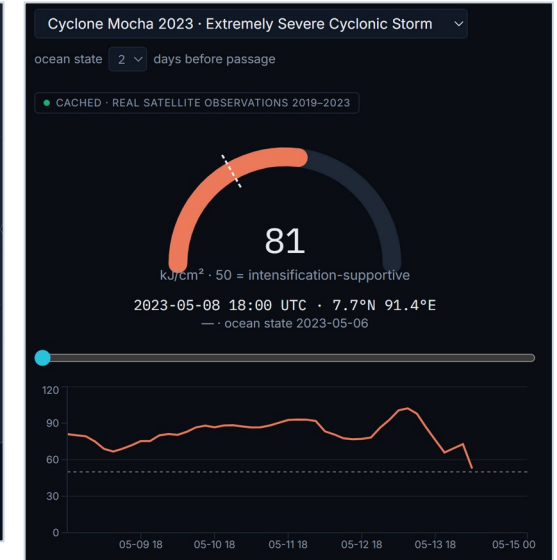
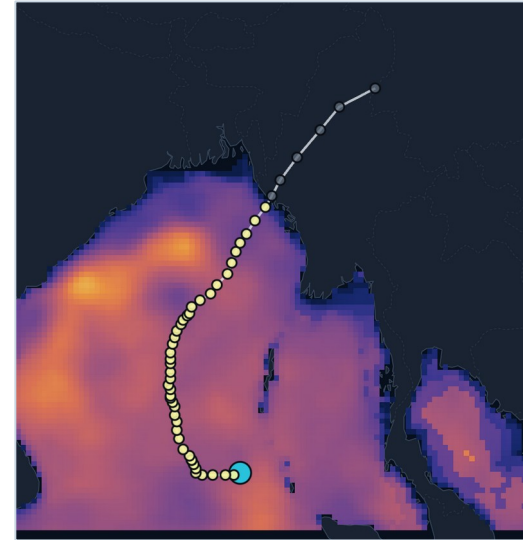
A continuous daily record of upper-ocean heat and stratification for the North Indian Ocean (2019–2023).

OPERATIONAL

Each profile ships with a calibrated error bar and the nearest independent Argo float — trust is visible, not assumed.

 **KEY OUTCOME**
full-basin subsurface view · daily · 0–1000 m · with measured confidence

Working prototype real event, held-out year



Cyclone Fuel Gauge — reconstructed TCHP along Cyclone Mocha's real IBTrACS track (May 2023, ocean state 2 days before passage): 52.7–102 kJ/cm² over the ocean, above the 50 kJ/cm² level commonly linked to intensification. Prototype output, not an operational forecast.

 **Click any cell**
full 0–1000 m profile + σ

 **Argo overlay**
nearest independent float

 **Export**
PDF / CSV report per point

 Details / Links of the reference and research work

#	Reference	Organisation	What it supports	Link
1	SIH26066 OceanEmbed problem statement	SIH · MoES / INCOIS	Scope, domain, 15 standard depths	sih.gov.in
2	INCOIS Tropical Cyclone Heat Potential service	INCOIS	Operational relevance of TCHP	incois.gov.in/site/services/tchp.jsp
3	GLORYS12V1 global ocean reanalysis (PS-named target)	Copernicus Marine	Training-target specification	doi.org/10.48670/moi-00021
4	Argo float profiles via argopy	Argo programme · Euro-Argo	Independent validation	argopy.readthedocs.io
5	NOAA OISST v2.1 · NCEI Blended Seawinds	NOAA NCEI	Satellite inputs used	ncei.noaa.gov
6	EN4 quality-controlled ocean analysis	UK Met Office Hadley Centre	Independent cross-check	metoffice.gov.uk/hadobs/en4
7	Bay of Bengal barrier layer (Vinayachandran et al., 2002)	JGR Oceans · AGU	Why salinity is an input	doi.org/10.1029/2001JC000831
8	Subsurface temperature reconstruction, ConvLSTM vs LightGBM (2022)	MDPI Remote Sensing	Model choice; thermocline is hardest	doi.org/10.3390/rs14133198
9	IBTrACS v04r01 best-track archive	NOAA NCEI	Real cyclone tracks	ncei.noaa.gov/products/international-best-track-archive


 Prototype source code & full research artifact: github.com/saad-46/oceanembed · all reported metrics are generated by the pipeline (docs/RESULTS.md)